

## SAFETY DATA SHEET

Revision Date 24-Dec-2021

Revision Number 4

### 1. Identification

**Product Name** 2,3-Dichlorobutane

**Cat No. :** AC113270000; AC113270250; AC113271000

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                     |            |
|---------------------|------------|
| Flammable liquids   | Category 2 |
| Acute oral toxicity | Category 4 |
| Skin Sensitization  | Category 1 |

#### Label Elements

**Signal Word**  
Danger

**Hazard Statements**  
Highly flammable liquid and vapor  
Harmful if swallowed  
May cause an allergic skin reaction

**Precautionary Statements****Prevention**

Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Avoid breathing dust/fume/gas/mist/vapors/spray  
Contaminated work clothing should not be allowed out of the workplace  
Keep away from heat/sparks/open flames/hot surfaces. - No smoking  
Keep container tightly closed  
Ground/bond container and receiving equipment  
Use explosion-proof electrical/ventilating/lighting equipment  
Use only non-sparking tools  
Take precautionary measures against static discharge  
Wear protective gloves/protective clothing/eye protection/face protection

**Skin**

If skin irritation or rash occurs: Get medical advice/attention  
Wash contaminated clothing before reuse  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower

**Ingestion**

IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell  
Rinse mouth

**Fire**

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

**Storage**

Store in a well-ventilated place. Keep cool

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

None identified

### 3. Composition/Information on Ingredients

| Component          | CAS No    | Weight % |
|--------------------|-----------|----------|
| 2,3-Dichlorobutane | 7581-97-7 | 98       |

### 4. First-aid measures

|                                            |                                                                                                                                                                                                                                                            |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                                                            |
| <b>Skin Contact</b>                        | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get medical attention.                                                                                                                               |
| <b>Inhalation</b>                          | Remove from exposure, lie down. Remove to fresh air.                                                                                                                                                                                                       |
| <b>Ingestion</b>                           | Clean mouth with water. Get medical attention.                                                                                                                                                                                                             |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. May cause allergic skin reaction. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Symptoms of allergic reaction may include rash, itching, swelling, trouble |

|                           |                                                                                                                                     |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Notes to Physician</b> | breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing<br>Treat symptomatically |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------|

## 5. Fire-fighting measures

|                                         |                                                                                                                   |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Carbon dioxide (CO <sub>2</sub> ). Dry chemical. Chemical foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                          |
| <b>Flash Point</b>                      | 18 °C / 64.4 °F                                                                                                   |
| <b>Method -</b>                         | No information available                                                                                          |
| <b>Autoignition Temperature</b>         | No information available                                                                                          |
| <b>Explosion Limits</b>                 |                                                                                                                   |
| <b>Upper</b>                            | No data available                                                                                                 |
| <b>Lower</b>                            | No data available                                                                                                 |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                          |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                          |

### Specific Hazards Arising from the Chemical

Flammable. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Vapors may form explosive mixtures with air.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen chloride gas.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
2

**Flammability**  
3

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                  |                                                                                        |
|----------------------------------|----------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Remove all sources of ignition. Take precautionary measures against static discharges. |
| <b>Environmental Precautions</b> | See Section 12 for additional Ecological Information.                                  |

|                                             |                                                                                                                                                                                                                                               |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Avoid contact with skin and eyes. Do not breathe mist/vapors/spray. Use spark-proof tools and explosion-proof equipment. Use only non-sparking tools. Keep away from open flames, hot surfaces and sources of ignition. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges. |
| <b>Storage.</b> | Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep away from heat, sparks and flame. Flammables area. Keep container tightly closed in a dry and well-ventilated place. Incompatible Materials. Strong oxidizing agents. Strong bases.                                                                                                                         |

## 8. Exposure controls / personal protection

|                            |                                                                                  |
|----------------------------|----------------------------------------------------------------------------------|
| <b>Exposure Guidelines</b> | This product does not contain any hazardous materials with occupational exposure |
|----------------------------|----------------------------------------------------------------------------------|

limits established by the region specific regulatory bodies.

**Engineering Measures** Use explosion-proof electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas.

#### Personal Protective Equipment

**Eye/face Protection** Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection** No protective equipment is needed under normal use conditions.

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                               |                                 |
|-----------------------------------------------|---------------------------------|
| <b>Physical State</b>                         | Liquid                          |
| <b>Appearance</b>                             | Light yellow                    |
| <b>Odor</b>                                   | No information available        |
| <b>Odor Threshold</b>                         | No information available        |
| <b>pH</b>                                     | No information available        |
| <b>Melting Point/Range</b>                    | -80 °C / -112 °F                |
| <b>Boiling Point/Range</b>                    | 117 - 119 °C / 242.6 - 246.2 °F |
| <b>Flash Point</b>                            | 18 °C / 64.4 °F                 |
| <b>Evaporation Rate</b>                       | No information available        |
| <b>Flammability (solid,gas)</b>               | Not applicable                  |
| <b>Flammability or explosive limits</b>       |                                 |
| Upper                                         | No data available               |
| Lower                                         | No data available               |
| <b>Vapor Pressure</b>                         | No information available        |
| <b>Vapor Density</b>                          | 4.4                             |
| <b>Specific Gravity</b>                       | 1.100                           |
| <b>Solubility</b>                             | No information available        |
| <b>Partition coefficient; n-octanol/water</b> | No data available               |
| <b>Autoignition Temperature</b>               | No information available        |
| <b>Decomposition Temperature</b>              | No information available        |
| <b>Viscosity</b>                              | No information available        |
| <b>Molecular Formula</b>                      | C4 H8 Cl2                       |
| <b>Molecular Weight</b>                       | 127.01                          |

## 10. Stability and reactivity

|                                         |                                                                                          |
|-----------------------------------------|------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                                               |
| <b>Stability</b>                        | Stable under normal conditions.                                                          |
| <b>Conditions to Avoid</b>              | Incompatible products. Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Strong bases                                                    |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Hydrogen chloride gas           |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                 |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                            |

## 11. Toxicological information

### Acute Toxicity

**Product Information** No acute toxicity information is available for this product

**Component Information**

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component          | CAS No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|--------------------|-----------|------------|------------|------------|------------|------------|
| 2,3-Dichlorobutane | 7581-97-7 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

**Persistence and Degradability** No information available

**Bioaccumulation/ Accumulation** No information available.

**Mobility** No information available.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No

UN1993

|                             |                           |
|-----------------------------|---------------------------|
| <b>Hazard Class</b>         | 3                         |
| <b>Packing Group</b>        | II                        |
| <b>TDG</b>                  |                           |
| <b>UN-No</b>                | UN1993                    |
| <b>Hazard Class</b>         | 3                         |
| <b>Packing Group</b>        | II                        |
| <b>IATA</b>                 |                           |
| <b>UN-No</b>                | UN1993                    |
| <b>Proper Shipping Name</b> | FLAMMABLE LIQUID, N.O.S.* |
| <b>Hazard Class</b>         | 3                         |
| <b>Packing Group</b>        | II                        |
| <b>IMDG/IMO</b>             |                           |
| <b>UN-No</b>                | UN1993                    |
| <b>Proper Shipping Name</b> | Flammable liquid, n.o.s.  |
| <b>Hazard Class</b>         | 3                         |
| <b>Packing Group</b>        | II                        |

## 15. Regulatory information

### United States of America Inventory

| Component          | CAS No    | TSCA | TSCA Inventory notification -<br>Active-Inactive | TSCA - EPA Regulatory<br>Flags |
|--------------------|-----------|------|--------------------------------------------------|--------------------------------|
| 2,3-Dichlorobutane | 7581-97-7 | X    | ACTIVE                                           | -                              |

#### Legend:

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

**TSCA 12(b)** - Notices of Export Not applicable

### International Inventories

Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component          | CAS No    | DSL | NDL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL |
|--------------------|-----------|-----|-----|-----------|-------|------|------|------|-------|------|
| 2,3-Dichlorobutane | 7581-97-7 | -   | X   | 231-486-4 | -     | X    | X    | X    | -     | -    |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### U.S. Federal Regulations

|                                                             |                                    |
|-------------------------------------------------------------|------------------------------------|
| <b>SARA 313</b>                                             | Not applicable                     |
| <b>SARA 311/312 Hazard Categories</b>                       | See section 2 for more information |
| <b>CWA (Clean Water Act)</b>                                | Not applicable                     |
| <b>Clean Air Act</b>                                        | Not applicable                     |
| <b>OSHA - Occupational Safety and Health Administration</b> | Not applicable                     |
| <b>CERCLA</b>                                               | Not applicable                     |

**California Proposition 65** This product does not contain any Proposition 65 chemicals.

### U.S. State Right-to-Know Regulations

| Component          | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|--------------------|---------------|------------|--------------|----------|--------------|
| 2,3-Dichlorobutane | X             | -          | X            | -        | -            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): N  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security** This product does not contain any DHS chemicals.

**Other International Regulations**

**Mexico - Grade** No information available

**Authorisation/Restrictions according to EU REACH****Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component          | CAS No    | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|--------------------|-----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| 2,3-Dichlorobutane | 7581-97-7 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component          | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|--------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| 2,3-Dichlorobutane | 7581-97-7 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y45                      |

**16. Other information**

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**Revision Summary** This document has been updated to comply with the US OSHA HazCom 2012 Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS).

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**